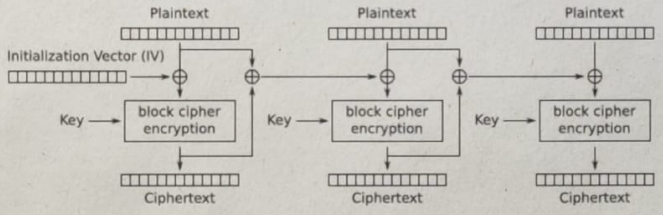
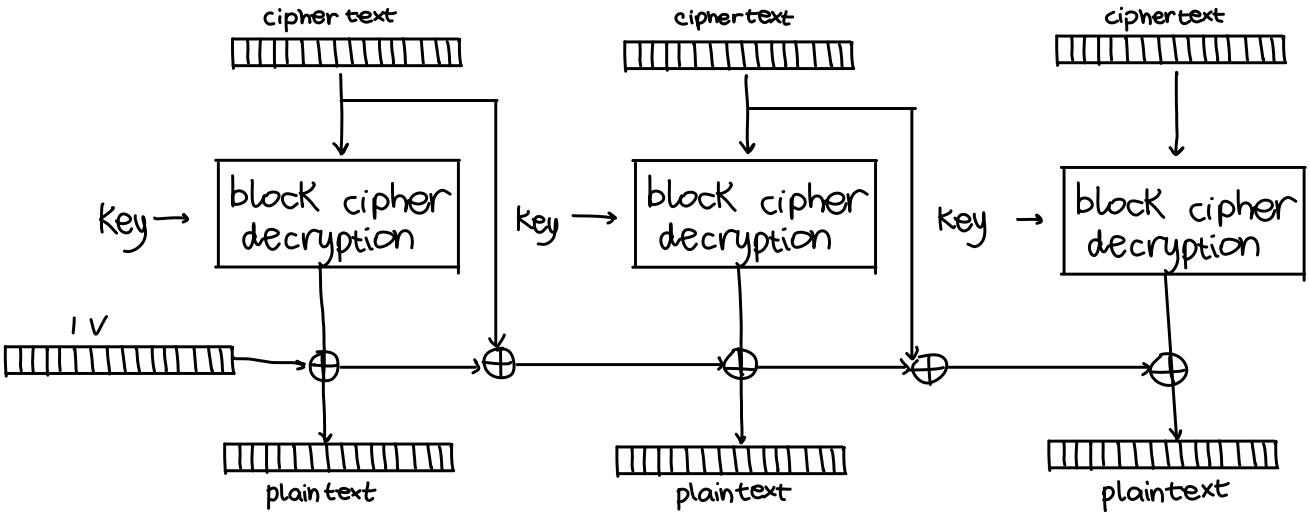
Febbraio 2025

1. Crittografia

The following encryption schema for *block encription* is used in Kerberos v.4.



Draw the corresponding encryption schema to be used for decryption.



2. Public Key Cryptography

Consider the Diffie-Hellman Key Exchange protocol with $q = 11$.

Note: You may find it useful to use the tabulation of the exponentiation function modulo 11, i.e. $f(x, y) = x^y \bmod 11$, which is provided below.

$x \backslash y$	1	2	3	4	5	6	7	8	9	10
2	2	4	8	5	10	9	7	3	6	1
3	3	9	5	4	1	3	9	5	4	1
4	4	5	9	3	1	4	5	9	3	1
5	5	3	4	9	1	5	3	4	9	1
6	6	3	7	9	10	5	8	4	2	1
7	7	5	2	3	10	4	6	9	8	1
8	8	9	6	4	10	3	2	5	7	1
9	9	4	3	5	1	9	4	3	5	1
10	10	1	10	1	10	1	10	1	10	1

(a) What are the possible values of the base α?

○ 1 ✗ 2 ○ 3 ○ 4 ○ 5 ✗ 6 ✗ 7 ✗ 8 ○ 9 ○ 10

Please explain why:

Perchè 2, 6, 7 e 8 sono radice primitiva di q=11, perchè i loro moduli ritornano tutti i numeri compresi tra 1 e q-1 (10) senza ripetizioni.

(b) You are Charlie. You know that Alice and Bob agreed to run the protocol with $q = 11$ and with α set to the largest among the possible values of α (see the previous question). You overhear a run of the protocol between Alice and Bob in which Alice sends the number 7 to Bob and she receives back the number 10. What is the key exchanged by Alice and Bob?

○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7 ○ 8 ○ 9 ✗ 10

Please explain why:

$Y_B = \alpha^{x_B} \bmod q \rightarrow 10 = 8^{x_B} \bmod 11$

$8^2 \bmod 11 = 64 \bmod 11 = 9$
$8^3 \bmod 11 = 9 \cdot 8 \bmod 11 = 72 \bmod 11 = 6$
$8^4 \bmod 11 = 6 \cdot 8 \bmod 11 = 48 \bmod 11 = 4$
$8^5 \bmod 11 = 4 \cdot 8 \bmod 11 = 32 \bmod 11 = 10$

$x_B = 5$

$K_B = Y_A^{x_B} \bmod q \rightarrow K_B = 7^5 \bmod 11$

$7^2 \bmod 11 = 49 \bmod 11 = 5$
$7^3 \bmod 11 = 5 \cdot 7 \bmod 11 = 35 \bmod 11 = 2$
$7^4 \bmod 11 = 2 \cdot 7 \bmod 11 = 14 \bmod 11 = 3$
$7^5 \bmod 11 = 3 \cdot 7 \bmod 11 = 21 \bmod 11 = 10$

$K = 10$

4. Secure Elements

(a) Which of the following functionalities can be provided by a *smartcard*?

✗ A. client-side support to the generation of digital siguatures
B. server-side support to the generation of digital siguatures
C. client-side support to the validation of digital siguatures
D. server-side support to the validation of digital siguatures
E. client-side support to the establishment of an SSL/TLS session
F. server-side support to the establishment of an SSL/TLS session

(b) Which of the following functionalities can be provided by a *hardware security module*?

A. client-side support to the generation of digital siguatures
✗ B. server-side support to the generation of digital siguatures
C. client-side support to the validation of digital siguatures
D. server-side support to the validation of digital siguatures
E. client-side support to the establishment of an SSL/TLS session
✗ F. server-side support to the establishment of an SSL/TLS session

3. **Security Protocols**

SuperMail wants every email to be authenticated and protected from modification or tampering while it is transit from the sender to the receiver. Suppose Alice is sending an email M to Bob. Given SuperMail's design constraints, which of the following options would be a secure way to protect the authenticity and integrity of her email?

(a) Alice's software should encrypt M under Bob's public key. In other words, Alice's software should send $E_{K_B}(M)$ to Bob.

Bob non può verificare l'integrità né autent. di M. M può venire anche sostituito da un altro messaggio visto che la cifratura avviene con la chiave pubblica di Bob, conosciuta da tutti per definizione.

(b) Alice's software should send M along with a digital signature on M using Alice's private key. In other words, Alice should send $M \| E(K_A^{-1}, h(M))$.

L'autenticità è garantita in quanto l'hash del messaggio viene cifrato con la chiave di A. L'integrità è garantita in quanto è presente l'hash, ossia il fingerprint, del messaggio, che non può essere modificato.

(c) Alice's software should choose a new symmetric key k for this email, send an encryption of k under Bob's public key, and also send an encryption of M under k using a stream cipher such as RC4. In other words, Alice should send $E(K_B, k) \| M \oplus RC4(k)$.

Non sono presenti identificativi o firme, quindi Bob non può sapere se M proviene da Alice

(d) Alice's software should choose a new symmetric key k for this email. Then it should send four pieces of information: the message M, a MAC on M under the key k, an encryption of k under Bob's public key, and a digital signature on k using Alice's private key. In other words, Alice should send $M \| MAC_k(M) \| E(K_B, k) \| E(K_A^{-1}, k)$.

La chiave k è visibile a tutti spacchettando l'ultima parte di messaggio con la chiave pubblica di Alice.
Bob può verificare se il messaggio è arrivato integro ma non può sapere se è stato modificato.